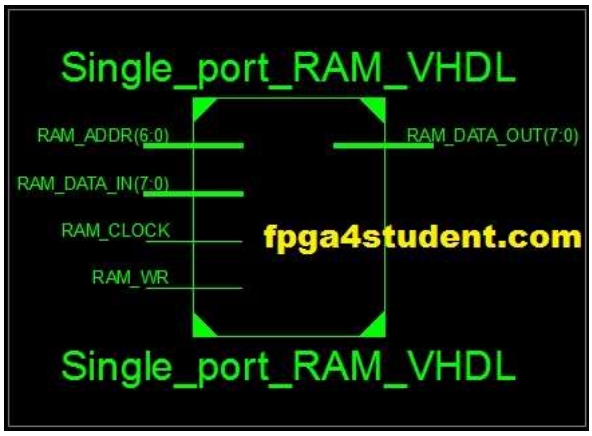
VHDL code for single-port RAM

This VHDL post presents a VHDL code for a single-port RAM (Random Access Memory). The VHDL testbench code is also provided to test the single-port RAM in Xilinx ISIM. The RAM's size is 128x8 bit.



As shown in the figure above, the 128x8 single port RAM in VHDL has following inputs and outputs:

1. RAM_CLOCK: the clock signal for sequentially writing data to the single-port RAM.

2. RAM_DATA_IN: 8-bit input data to be written to RAM at the provided input address RAM_ADDR when it is enabled.

3. RAM_WR: Write enable signal for writing to RAM, only if RAM_WR = 1, RAM_DATA_IN is written to the RAM at the rising edge of the clock signal.

4. RAM_ADDR: 6-bit Address where 8-bit input data are written to and data are read out.

5. RAM_DATA_OUT: 8-bit output data read out from the provided input address RAM_ADDR.

VHDL code for the single-port RAM:

```
-- fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL project: VHDL code for a single-port RAM
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
USE ieee.numeric_std.ALL;

-- A 128x8 single-port RAM in VHDL
entity Single_port_RAM_VHDL is
port(
 RAM_ADDR: in std_logic_vector(6 downto 0); -- Address to write/read RAM
 RAM_DATA_IN: in std_logic_vector(7 downto 0); -- Data to write into RAM
 RAM_WR: in std_logic; -- Write enable
 RAM_CLOCK: in std_logic; -- clock input for RAM
 RAM_DATA_OUT: out std_logic_vector(7 downto 0) -- Data output of RAM
);
end Single_port_RAM_VHDL;

architecture Behavioral of Single_port_RAM_VHDL is
-- define the new type for the 128x8 RAM
type RAM_ARRAY is array (0 to 127 ) of std_logic_vector (7 downto 0);
-- initial values in the RAM
signal RAM: RAM_ARRAY :=(
   x"55",x"66",x"77",x"67",-- 0x00:
   x"99",x"00",x"00",x"11",-- 0x04:
   x"00",x"00",x"00",x"00",-- 0x08:
   x"00",x"00",x"00",x"00",-- 0x0C:
   x"00",x"00",x"00",x"00",-- 0x10:
   x"00",x"00",x"00",x"00",-- 0x14:
   x"00",x"00",x"00",x"00",-- 0x18:
   x"00",x"00",x"00",x"00",-- 0x1C:
```

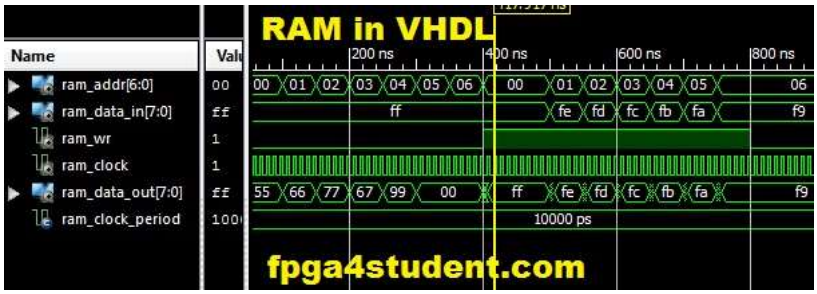
```
   x"00",x"00",x"00",x"00",-- 0x20:
   x"00",x"00",x"00",x"00",-- 0x24:
   x"00",x"00",x"00",x"00",-- 0x28:
   x"00",x"00",x"00",x"00",-- 0x2C:
   x"00",x"00",x"00",x"00",-- 0x30:
   x"00",x"00",x"00",x"00",-- 0x34:
   x"00",x"00",x"00",x"00",-- 0x38:
   x"00",x"00",x"00",x"00",-- 0x3C:
   x"00",x"00",x"00",x"00",-- 0x40:
   x"00",x"00",x"00",x"00",-- 0x44:
   x"00",x"00",x"00",x"00",-- 0x48:
   x"00",x"00",x"00",x"00",-- 0x4C:
   x"00",x"00",x"00",x"00",-- 0x50:
   x"00",x"00",x"00",x"00",-- 0x54:
   x"00",x"00",x"00",x"00",-- 0x58:
   x"00",x"00",x"00",x"00",-- 0x5C:
   x"00",x"00",x"00",x"00",
   x"00",x"00",x"00",x"00",
   x"00",x"00",x"00",x"00",
   x"00",x"00",x"00",x"00",
   x"00",x"00",x"00",x"00",
   x"00",x"00",x"00",x"00",
   x"00",x"00",x"00",x"00",
   x"00",x"00",x"00",x"00"
   );
begin
process(RAM_CLOCK)
begin
 if(rising_edge(RAM_CLOCK)) then
 if(RAM_WR='1') then -- when write enable = 1,
 -- write input data into RAM at the provided address
 RAM(to_integer(unsigned(RAM_ADDR))) <= RAM_DATA_IN;
 -- The index of the RAM array type needs to be integer so
 -- converts RAM_ADDR from std_logic_vector -> Unsigned -> Interger using numeric_
 end if;
 end if;
end process;
 -- Data to be read out
 RAM_DATA_OUT <= RAM(to_integer(unsigned(RAM_ADDR)));
end Behavioral;
```

VHDL Testbench code for the single-port RAM:

```
-- fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL project: VHDL code for a single-port RAM
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
USE ieee.std_logic_unsigned.ALL;
-- VHDL testbench code for the single-port RAM
ENTITY tb_RAM_VHDL IS
END tb_RAM_VHDL;

ARCHITECTURE behavior OF tb_RAM_VHDL IS

    -- Component Declaration for the single-port RAM in VHDL

    COMPONENT Single_port_RAM_VHDL
    PORT(
         RAM_ADDR : IN  std_logic_vector(6 downto 0);
         RAM_DATA_IN : IN  std_logic_vector(7 downto 0);
         RAM_WR : IN  std_logic;
         RAM_CLOCK : IN  std_logic;
         RAM_DATA_OUT : OUT  std_logic_vector(7 downto 0)
        );
```

```
    END COMPONENT;


   --Inputs
   signal RAM_ADDR : std_logic_vector(6 downto 0) := (others => '0');
   signal RAM_DATA_IN : std_logic_vector(7 downto 0) := (others => '0');
   signal RAM_WR : std_logic := '0';
   signal RAM_CLOCK : std_logic := '0';

  --Outputs
   signal RAM_DATA_OUT : std_logic_vector(7 downto 0);

   -- Clock period definitions
   constant RAM_CLOCK_period : time := 10 ns;

BEGIN

 -- Instantiate the single-port RAM in VHDL
   uut: Single_port_RAM_VHDL PORT MAP (
          RAM_ADDR => RAM_ADDR,
          RAM_DATA_IN => RAM_DATA_IN,
          RAM_WR => RAM_WR,
          RAM_CLOCK => RAM_CLOCK,
          RAM_DATA_OUT => RAM_DATA_OUT
        );

   -- Clock process definitions
   RAM_CLOCK_process :process
   begin
  RAM_CLOCK <= '0';
  wait for RAM_CLOCK_period/2;
  RAM_CLOCK <= '1';
  wait for RAM_CLOCK_period/2;
   end process;

   stim_proc: process
   begin
  RAM_WR <= '0';
  RAM_ADDR <= "0000000";
  RAM_DATA_IN <= x"FF";
      wait for 100 ns;
  -- start reading data from RAM
  for i in 0 to 5 loop
  RAM_ADDR <= RAM_ADDR + "0000001";
      wait for RAM_CLOCK_period*5;
  end loop;
  RAM_ADDR <= "0000000";
  RAM_WR <= '1';
  -- start writing to RAM
  wait for 100 ns;
  for i in 0 to 5 loop
  RAM_ADDR <= RAM_ADDR + "0000001";
  RAM_DATA_IN <= RAM_DATA_IN-x"01";
      wait for RAM_CLOCK_period*5;
  end loop;
  RAM_WR <= '0';
      wait;
   end process;

END;
```

Simulation waveform for the single-port RAM in VHDL:



The simulation waveform shows the correct writing and reading operation of the single-port RAM in VHDL. The VHDL code for the single-port RAM is used for the 8-bit Microcontroller here.

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